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TOP TIER
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Portable Generator Homeowner Manual

Plain-English guidance for safe, simple portable generator use during a power outage

Important
This is a use guide, not an installation guide.
Only connect a generator to your home if you already have a listed interlock or a listed manual transfer switch installed for that panel.
If you do not have one, power appliances directly with extension cords only.

1. Safety Rules That Never Change

- Run the generator outdoors only. Keep it away from doors, windows, vents, and the garage.
- Use carbon monoxide alarms in the home. Generator exhaust can be deadly.
- Keep the generator dry. Do not run it in standing water or where rain can hit the controls.
- Never backfeed through a dryer outlet, range outlet, or any receptacle.
- Never refuel a hot generator. Let it cool first.
- Do not change neutral, bonding, grounding, or wiring unless a licensed electrician tells you to.

2. Know Your Setup

Interlock system: A sliding or blocking device at the main panel prevents the main breaker and generator breaker from being on at the same time.

Manual transfer switch: A separate switch box lets you move selected circuits from utility power to generator power.

If you are not sure which one you have, stop and identify it before the next outage.

3. Make a Simple Outage Plan

Before a storm, decide what must stay on and what must stay off.

Priority	Examples
Must stay on	Refrigerator, freezer, furnace blower, boiler controls, sump pump, well pump, medical equipment, a few lights, router
Nice to have	Microwave, coffee maker, TV, garage door opener
Leave off	Electric water heater, dryer, range, central AC, EV charger, hot tub, pool equipment

Using an Interlock System

Use these steps only if your home already has a listed interlock, inlet, and generator breaker.

4. Before Starting the Generator

- Move the generator outdoors and place it on a firm, level surface.
- Check fuel, oil, and the owner's manual for your model.
- Turn off or unplug major loads inside the home.
- At the panel, turn off large non-emergency breakers first: water heater, dryer, range, central AC, EV charger, and pool equipment.
- If you use a 'dark panel' start, turn branch breakers off and plan to turn back on only what you need.

5. Transfer the House to Generator Power

- Make sure the generator breaker in the panel is OFF.
- Plug the generator cord into the inlet box and then into the generator.
- Start the generator with no house load on it and let it stabilize for about 30 to 60 seconds.
- Turn OFF the main breaker. This disconnects the house from utility power.
- Move the interlock so the generator breaker can be turned ON.
- Turn ON the generator breaker.
- Bring house circuits on one at a time. Start with the most important loads first.

Good breaker order

1. Refrigerator / freezer
2. Furnace or boiler controls
3. Sump pump or well pump if needed
4. A few lighting circuits
5. Router / modem / charging
6. Microwave or coffee maker, one at a time

Wait a few seconds between motor loads. If the engine bogs down, lights dip hard, or the generator trips, turn off the last load you added.

6. While the Generator Is Running

- Keep non-essential breakers off.
- Do not run several heavy appliances at the same time.
- Avoid starting two motor loads together, such as a refrigerator and a sump pump.
- Check fuel level before it gets low. Plan refueling so the unit can cool down first.

7. Return to Utility Power

- Turn off branch circuits that were being powered by the generator.
- Turn OFF the generator breaker.
- Move the interlock back to the utility position.
- Turn ON the main breaker once utility power is stable.
- Turn branch circuits back on in a controlled order, then let the generator run briefly with little or no load if your manual recommends it before shutting it down.

Using a Manual Transfer Switch

Use these steps only if your home has a listed manual transfer switch already installed.

8. Before Starting

- Move the generator outdoors and check fuel and oil.
- Review which circuits are connected to the transfer switch.
- Turn off large loads on those circuits before starting.
- Set transfer switch handles or toggles to LINE, OFF, or the position your label shows before startup.

9. Transfer Selected Circuits to Generator Power

- Plug the generator into the inlet or transfer switch connection point.
- Start the generator and let it stabilize.
- Move transfer switch circuits to GENERATOR one at a time.
- Start with critical loads first: refrigerator, heating controls, sump or well pump, lights, and communications.
- Add convenience loads only if the generator is handling the essential loads easily.

10. Good Load Management

Portable generators do not like surprise load spikes. The safest habit is to run essentials steadily and rotate short-use appliances.

Keep on most of the time	Use only when needed	Usually leave off
Fridge, freezer, small lights, router, furnace controls, sump pump if needed	Microwave, coffee maker, toaster, garage door opener	Water heater, dryer, range, central AC, EV charger, hot tub

11. Return to Utility Power

- Move transfer switch circuits back to LINE or OFF, following the labels on your switch.
- Shut down the generator using the steps in the owner's manual.
- Store the cord safely and refuel only after the generator has cooled.

12. Universal Rules for Both Systems

- Start with fewer loads than you think you can run.
- Motors take extra power to start, even when their normal running wattage is modest.
- One heavy kitchen appliance at a time is usually the smart rule.
- If the generator is struggling, remove load before it trips.

Wattage, Load Limits, and Troubleshooting

Use this page to make fast decisions during an outage.

13. Running Watts vs Starting Watts

Running watts are the power needed after an appliance is already operating. **Starting watts** are the extra power needed when a motor first starts. Refrigerators, sump pumps, well pumps, and furnace blowers often need more power for a few seconds at startup.

14. Typical Household Loads

Appliance	Typical running watts	What to know
Refrigerator / freezer	150 to 700	Usually has a higher startup surge
Furnace blower	400 to 800	Startup can be much higher than running load
Sump pump	800 to 1,050	Start it by itself if possible
Well pump	1,000+	Often one of the biggest surge loads in the house
Microwave	1,000 to 1,500	Use one at a time with other heavy loads off
Coffee maker / toaster	800 to 1,500	Small generators feel these loads quickly
Electric water heater	4,500	Usually left off on portable generator power
Dryer	5,400+	Usually left off on portable generator power

These are planning numbers only. Check the data plate on your actual equipment when you can.

15. What Smaller and Larger Portable Generators Usually Handle

Generator size	Usually reasonable	Usually not reasonable
Around 3,000W	Fridge, lights, router, charging, maybe one small appliance at a time	Water heater, dryer, most well pumps, multiple heavy appliances
Around 5,000W	Fridge, furnace blower, lights, router, some sump pumps, microwave if managed carefully	Water heater, dryer, stacking several large loads
Around 7,500W to 10,000W	Most essential circuits, with careful sequencing	Running the whole house as if utility power never left

16. If Something Goes Wrong

- Generator bogs down or lights dim hard: turn off the last load you added.

- Generator breaker trips: remove load, reset if the manual allows it, and restart with fewer circuits.

- A pump or refrigerator will not start: turn off other loads and try again with that load by itself.

- No power at the panel: check cord connection, main breaker position, interlock position, and generator breaker position.

- GFCI trips or cords get warm: stop and inspect for damage, moisture, or overload.

Need help? / One last reminder
Top Tier Electrical Services can help with properly installed inlets, interlocks, transfer switches, generator-ready panels, troubleshooting, and service calls. Call 616.334.7159 or visit TopTier-Electrical.com.

Portable generators are limited power sources. Safe use comes down to three habits: isolate from utility power correctly, add load in a controlled order, and shed load early if the generator struggles.